Quarry: Product Requirements Document

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Product	Quarry, AI-assisted B2B lead generation
Document	Product Requirements Document (PRD)
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> **How to read this document.** Statements about what the product does come from the current code and are marked *Built*. Statements about users, market and pricing are *hypotheses*: the concierge-pilot design doc records that no customer conversations have happened yet, so nothing here claims demand has been proven. Competitor facts come from `reports/Quarry next feature roadmap.md`, which draws on vendor-authored sources and is directional only.

1. Summary

Quarry turns a plain-language description of an ideal customer into a reviewed list of matching companies, decision makers and verified email addresses, then helps the user act on that list: score it, watch it for buying signals, send sequenced outreach, and push it into a CRM. Every lead carries its source and a checked-at date, and every personal record can be erased or suppressed.

The product bets on **data trust**. Incumbents lose users to inaccurate contacts, opaque credit pricing and auto-renew lock-in. Quarry's answer is honest email labels (Verified, Likely, Risky, Invalid), per-field provenance, a cost estimate before any spend, and a compliance layer built in from the start.

1.1 One-paragraph pitch

”Describe who you sell to. Quarry searches the live web, builds a scored list of companies and the people who decide, tells you how sure it is about each email and where each fact came from, and lets you send to them from your own mailbox or hand them to your CRM, without surprise bills and without leaving a trail of personal data you cannot delete.”

2. Problem and opportunity

2.1 The problem

People who build B2B prospect lists (agency owners, freelance SDRs, founders doing their own outbound) face four recurring problems:

1. **Stale or wrong data.** Contact databases go out of date, and bounce rates of 20 to 30% are reported for some vendors (vendor-sourced, directional). Bad lists burn sender reputation.
2. **Opaque, expensive pricing.** Credit systems are hard to predict; enterprise databases are quote-only; entry tools sit around $50 to $120 per month and agencies at $2,500 and up. There is a gap between them.
3. **Fragmented workflow.** Finding companies, finding people, verifying emails, sending mail and syncing to a CRM are usually four or five tools.
4. **Compliance exposure.** CAN-SPAM, CASL, UK PECR, GDPR and India’s DPDP Act all touch cold outreach, and most small teams have no suppression list, no source record and no way to honour an erasure request.

2.2 Why Quarry

Competitor layer	Examples	Gap Quarry targets
Contact databases	ZoomInfo, Apollo, Lusha, Seamless.AI	Fresh data found live from the web per ICP; honest accuracy labels
Enrichment engines	Clay	Guided, opinionated front end for non-technical sellers
Cold-email senders	Instantly, Smartlead, lemlist	Quarry’s edge is upstream of the send, but sending is included
AI SDR platforms	11x, Artisan	Transparent per-request pricing, reviewable output, no autonomous sending

The differentiator is not a single feature. It is the combination of **live discovery, visible provenance, honest email status, built-in compliance and predictable cost**.

3. Goals and non-goals

3.1 Goals

ID	Goal	Measure
G1	A first usable lead list within minutes of describing the ICP	Median time from launch to Completed (to be instrumented)
G2	Users can trust the data	Share of exported emails labelled Verified or Likely; bounce rate on a labelled 100 to 200 lead set
G3	No surprise spend	Cost estimate shown before every run; credits and quota visible in the product
G4	Safe by default	Zero cross-tenant data access; SSRF, CSRF and auth controls tested
G5	Compliance is a feature	Suppression honoured before enrichment, export and send; erasure completes end to end
G6	Close the loop	Users can send sequences and sync to a CRM without leaving Quarry

3.2 Non-goals (this release)

- A shared, resold contact database. Quarry processes data per customer; the default is not to build one.
- Autonomous sending without user review.
- Scraping logged-in LinkedIn sessions. LinkedIn-derived data comes from public search results only.
- Gmail or Microsoft OAuth mailbox access (restricted scopes trigger a security assessment). SMTP and IMAP are used instead.
- Postgres and horizontal scaling (documented as the next infrastructure step).
- Phone-number data and dialers.

- Native mobile apps and a browser extension.

4. Users and personas

> These personas are working hypotheses from the pilot design doc. They must be validated by interviews before they drive pricing or positioning.

Persona	Context	Job to be done	What they care about
Freelance SDR / lead-list builder	Builds lists for clients, paid per list	Produce a clean, defensible list fast	Accuracy, speed, CSV export, per-list cost
Agency owner	Runs outbound for several clients	Repeatable process across clients, team access	Seats and roles, shared lists, CRM push
Founder doing outbound	Sells their own product, no sales team	Find the 50 right companies and email them this week	Simplicity, sending from own mailbox, low cost
Team admin (secondary)	Owns a workspace	Control access, spend and compliance	Roles, billing, audit log, suppression

4.1 Roles inside a workspace

Role	Can do
Owner	Everything, including billing and ownership of the workspace
Admin	Manage members, invites, billing top-ups, integrations
Member	Create requests, lists, sequences; use the product day to day
Viewer	Read shared requests and lists

Role order is `owner > admin > member > viewer`; API dependencies enforce a minimum role per endpoint.

5. Product scope and requirements

Priority key: **P0** launch-critical, **P1** important, **P2** later. Status: **Built** is implemented and tested; **Partial** means implemented with a known limit; **Open** is not done.

5.1 Accounts and access (FR-A)

ID	Requirement	Pri	Status
FR-A1	Email and password registration and login	P0	Built
FR-A2	Google sign-in via a server-side redirect flow	P1	Built
FR-A3	Password reset by email link	P0	Built
FR-A4	HttpOnly cookie sessions with rotating refresh tokens and CSRF protection on writes	P0	Built
FR-A5	Bearer-token access for API clients	P1	Built
FR-A6	Rate limiting on login, register and reset; uniform error messages	P0	Built
FR-A7	Delete account and cascade-delete the user’s personal data	P0	Built

Acceptance (FR-A4). A cookie-authenticated POST without a matching `X-CSRF-Token` returns 403. A refresh token presented as an access token is rejected. Logout deletes the server-side session and clears cookies.

5.2 Lead requests: the core flow (FR-R)

ID	Requirement	Pri	Status
FR-R1	Guided wizard to define an ICP, optionally seeded from pasted text, URLs or files	P0	Built
FR-R2	LLM helpers: generate ICP, check consistency, suggest chips, find competitors	P1	Built
FR-R3	Launch a discovery run: queries, search, scrape, extract, score, enrich	P0	Built
FR-R4	”Direct companies” run: user supplies a company list or CSV; Quarry enriches it	P1	Built
FR-R5	Runs are asynchronous; launching returns a request id at once; progress by polling or server-sent events	P0	Built
FR-R6	Runs survive a server restart and can be cancelled	P0	Built
FR-R7	A run that fails keeps the partial leads and shows a failure reason	P0	Built
FR-R8	Cost and time estimate before launch, and a small sample run	P1	Built
FR-R9	Pipeline finds decision makers and competitors in parallel with leads	P1	Built
FR-R10	Optional ICP score filter (off by default so nothing is silently discarded)	P2	Built
FR-R11	Retry a lead whose AI enrichment failed	P1	Built
FR-R12	Request quota check and credit debit are atomic	P0	Built

Acceptance (FR-R7). If extraction fails for a page, the lead is kept, named from the page title, flagged `enrichment_failed` with a user-facing reason, and shown with a Retry action. It is never dropped silently.

5.3 Lead data, scoring and trust (FR-L)

ID	Requirement	Pri	Status
FR-L1	Company profile: industry, size, location, tech stack, infrastructure, funding, news, jobs, social signals	P0	Built
FR-L2	Match score 0 to 100 against the ICP	P0	Built
FR-L3	Intent score from four pillars (firmographic 25, technographic 20, intent 35, engagement 20) with confidence weights and a 15% per 30 days decay; bands HOT, WARM, NURTURE, WATCH, COLD	P1	Built
FR-L4	Four-state email status: Verified, Likely, Risky, Invalid, with reason, method and checked-at timestamp	P0	Built
FR-L5	Pluggable verifier with a bring-your-own provider key; catch-all awareness	P1	Built
FR-L6	Per-field provenance: source URL and collection date, shown in the UI	P0	Built
FR-L7	Region flag per lead (EU, UK, CA, IN) with a compliance note	P1	Built
FR-L8	Waterfall email enrichment across scrape, Hunter, Snov and People Data Labs, charged per success	P1	Built
FR-L9	Batch AI scoring job	P2	Built
FR-L10	Competitor discovery and comparison for the user’s own product	P2	Built

5.4 Lists, export and sharing (FR-X)

ID	Requirement	Pri	Status
FR-X1	Star, exclude and validate companies; add to named lead lists	P0	Built
FR-X2	Export companies, competitors, decision makers and lists as CSV, XLSX or PDF	P0	Built
FR-X3	CSV carries a source-notice line and opt-out instruction	P1	Built
FR-X4	Share requests and lists with a workspace (read access for teammates)	P1	Built
FR-X5	Suppressed contacts never appear in an export	P0	Built

5.5 Outreach (FR-O)

ID	Requirement	Pri	Status
FR-O1	Connect a mailbox over SMTP and IMAP; credentials encrypted at rest	P0	Built
FR-O2	Test the connection (SMTP login, IMAP login) before saving	P1	Built
FR-O3	Multi-step sequences, up to 8 steps, with merge variables and a preview	P0	Built
FR-O4	Send windows by timezone (default 09:00 to 17:00), daily caps per mailbox (default 40), and a warmup ramp from 5 per day growing 30% a day up to the cap	P0	Built
FR-O5	Reply and bounce detection by IMAP polling, threaded by Message-ID; DSN parsing; auto-reply detection	P0	Built
FR-O6	Auto-pause a mailbox when its hard-bounce rate passes 2% (after at least 20 sends; both configurable); resume manually	P0	Built
FR-O7	One-click unsubscribe headers and a footer with unsubscribe link and sender postal address	P0	Built
FR-O8	DNS preflight: SPF, DKIM, DMARC check for the sending domain	P1	Built
FR-O9	Suppressed addresses are skipped at send time	P0	Built
FR-O10	Sequence stats: sent, replied, bounced (reply rate is the headline; open tracking is secondary)	P1	Built
FR-O11	Gmail and Microsoft OAuth mailboxes	P2	Open (out of scope)

5.6 Signals and watch (FR-W)

ID	Requirement	Pri	Status
FR-W1	Saved searches that re-run on a schedule	P1	Built
FR-W2	Signal events (hiring, funding, news, tech-stack change) with snapshots to detect change	P1	Built
FR-W3	Per-lead signal feed and an account-wide signal list	P1	Built
FR-W4	Email and Slack alerts when a saved search finds new matches or signals; alerts are seeded so existing matches are not announced	P1	Built

5.7 Integrations (FR-I)

ID	Requirement	Pri	Status
FR-I1	Public API keys, created and revoked in the UI	P1	Built
FR-I2	Outbound webhooks with delivery log and test button	P1	Built
FR-I3	HubSpot connection (OAuth) and list push, run as a background job with per-lead sync status	P1	Built
FR-I4	CSV export endpoint for automation tools	P1	Built
FR-I5	Salesforce, Pipedrive, Zapier and Make connectors	P2	Open

5.8 Workspaces, billing and usage (FR-B)

ID	Requirement	Pri	Status
FR-B1	Organizations with owner, admin, member and viewer roles; email invites	P0	Built
FR-B2	Plans: Free, Pro, Team with request, credit, seat and CRM-connection limits	P0	Built
FR-B3	Credit ledger: monthly grant, spend, top-ups; two separate balances (request credits, email-lookup credits)	P0	Built
FR-B4	Payment providers behind one interface: Stripe, Razorpay, and a mock for local demos	P1	Built
FR-B5	Credit top-up packs, credited only after a verified paid webhook, idempotent per event	P0	Built
FR-B6	Usage summary: requests, spend and quota remaining	P1	Built

PLAN CATALOGUE (FROM `BILLING/PLANS.PY`)

Plan	Price / month	Requests / month	Credits / month	Seats	CRM connections
Free	$0	3	25	3	0
Pro	$49	50	500	5	1
Team	$199	250	2,500	25	5

Top-up packs: 100 credits for $10, 500 for $40, 2,000 for $140. A lead request costs 1 credit. These prices are placeholders set in code and **have not been validated with customers**.

5.9 Compliance and trust (FR-C)

ID	Requirement	Pri	Status
FR-C1	Workspace-wide suppression ledger checked before enrichment, export and send	P0	Built
FR-C2	Public opt-out page and signed opt-out links	P0	Built
FR-C3	Person-level erasure tool	P0	Built
FR-C4	Audit log of compliance-relevant actions	P0	Built
FR-C5	30-day retention purge for rejected and unpromoted leads	P1	Built
FR-C6	Reject and restore a lead	P1	Built
FR-C7	Privacy notice page	P0	Built
FR-C8	Legal review of data-broker status, legitimate-interest assessment, DPA and subprocessor list	P0	**Open: needs counsel**

6. User journeys

6.1 Journey A: first list in one session

1. Sign up (email or Google).
2. Create a request: describe the product and buyer. Optionally paste a URL or document to seed the ICP.
3. Review and edit the generated ICP in the wizard.
4. See a cost and time estimate. Optionally run a small sample first.
5. Launch. The request page shows live progress; the user can leave and return.
6. Review leads: match score, tech stack, signals, decision makers, email status with source and date.
7. Star, exclude or reject leads; add keepers to a list.
8. Export a CSV, or continue to journey B.

6.2 Journey B: from list to conversation

1. Connect a mailbox; Quarry tests the SMTP and IMAP login.
2. Run the DNS preflight for the sending domain.
3. Build a sequence (up to 8 steps) with merge variables; preview it against a real contact.
4. Enroll a list. Quarry sends inside the send window, under the daily cap, skipping suppressed addresses.
5. Replies and bounces are detected by IMAP polling; replying contacts leave the sequence.
6. Optionally push the list to HubSpot and watch per-lead sync status.

6.3 Journey C: a recipient opts out

1. The recipient clicks Unsubscribe in the footer or the one-click header.
2. A signed token adds them to the workspace suppression ledger and records an audit entry.
3. Future enrichment, export and sends skip them. They can be erased entirely by an admin.

7. Non-functional requirements

Area	Requirement	Current state
Security	Object-level authorization on every resource; no header identity shortcut	Built, tested
	SSRF protection on every outbound fetch of a user-influenced URL	Built; DNS-pinned connections for httpx and requests; Playwright checked but not pinned (off by default)
	Secrets encrypted at rest; no secrets in job payloads	Built
Reliability	Durable jobs, retry with backoff, recovery after restart	Built
	Single writer: one API worker only	Documented limit
Performance	A run returns a request id immediately; long work is off the request path	Built. Run duration is **not yet measured**; the “about 6 minutes” figure in early mockups is an estimate
Scalability	Single instance on SQLite	Fine for a pilot; Postgres is the scale-out path
Privacy	Third-party personal data deletable by request, lead, account	Built
Accessibility	Keyboard focus trap and Escape in dialogs, labelled controls, status never conveyed by colour alone	Built; formal contrast audit pending
Browser support	Current evergreen browsers; responsive to phone width	Built
Observability	Structured logs; failure reasons surfaced to the user; audit log	Built; no metrics or tracing backend yet

Area	Requirement	Current state
Maintainability	CI runs ruff, pytest, eslint, vitest and a production build	Built

8. Success metrics

Metric	Why	Target (proposed, to calibrate)
Activation: users who launch a run within 24 hours of signup	Product clarity	Set a baseline first
Time to first result	Core promise	Under 10 minutes for a 25-lead run
Email accuracy: bounce rate on delivered lists	Trust claim	Under 2% hard bounce on Verified+Likely
Enrichment failure rate	Pipeline health	Under 5% of leads flagged `enrichment_failed`
Run success rate	Reliability	Over 95% of runs Completed
Reply rate on sequences	Outcome that matters	Track per user; no universal target
Opt-out handled	Compliance	100% of suppressions honoured at the next send
Paid conversion	Business	Unknown until pricing is validated

Open measurement gap: Quarry cannot see bounces or replies for mail sent outside Quarry, and the accuracy of the LLM-extracted contacts is unmeasured. The roadmap proposes a labelled set of 100 to 200 leads.

9. Constraints, assumptions and risks

9.1 Constraints

- Solo builder; free tiers and bring-your-own keys are the build-time cost model.

- One process and one SQLite writer: the job queue, schedulers and write lock are in-process.
- No mailbox OAuth, to avoid restricted-scope security assessments.

9.2 Assumptions (unvalidated)

1. A niche exists where live-web data beats stale databases.
2. Buyers will pay per request or per month for transparent, reviewed lists.
3. Users accept bringing their own Serper key, or an operator-paid key at a marked-up credit price.

9.3 Risk register

#	Risk	Likelihood	Impact	Mitigation
1	No demand evidence (no customer conversations yet)	High	High	Run the interview and concierge plan before more features; time-boxed go/no-go
2	Data-broker or privacy-law exposure from storing third-party contacts	Medium	High	Per-customer processing default; suppression, erasure, retention; **legal review before hosted launch**
3	LLM extraction errors or hallucinated contacts	Medium	High	Provenance and checked-at on every field; verification statuses; measure on a labelled set
4	Deliverability harm to customers' domains	Medium	High	Caps, warmup, auto-pause, unsubscribe headers, DNS preflight
5	SSRF or abuse via scraped URLs once hosted	Low after fixes	High	URL guard with DNS pinning; residual Playwright gap documented
6	SQLite single-writer limits growth	Medium	Medium	One worker today; Postgres migration is the next infrastructure step
7	Upstream cost spikes (Serper, LLM)	Medium	Medium	Estimates, quotas, credits, bring-your-own keys
8	Unvalidated pricing	High	Medium	Treat plan prices as placeholders until interviews

10. Release plan and open items

All four roadmap phases (foundation, data trust and compliance, features, growth) are implemented. What remains before a hosted launch:

Item	Type	Priority
Legal review: data-broker status, legitimate-interest assessment, DPA, subprocessors	Compliance	P0
Exercise real services end to end (full pipeline with a live Serper key, email verification providers, SMTP and IMAP, Stripe, Razorpay, HubSpot)	Verification	P0
Re-check shared work with two accounts in one organization	QA	P3
Response-size cap in the scraper	Hardening	P3
Customer interviews and a concierge pilot to validate persona, niche and price	Discovery	P0
Postgres and multi-worker queue	Scale	P2
Salesforce, Pipedrive, Zapier and Make connectors; browser extension	Growth	P2

11. Glossary

Term	Meaning
ICP	Ideal Customer Profile: the description of the company and buyer you want
Request	One discovery run and its results
Lead	A matched company with its enrichment
Decision maker	A person at a lead who is likely to buy
Provenance	Where a fact came from (URL) and when it was collected
Suppression	A do-not-contact entry that blocks enrichment, export and sending
Waterfall	Trying several data sources in order and paying only for a hit
Warmup	A gradual increase in daily send volume for a new mailbox
DSN	Delivery Status Notification: the bounce message mail servers return
SSRF	Server-side request forgery: tricking the server into fetching internal addresses